101 C++ PROGRAMS - COMPLETE COLLECTION

1. CALCULATOR - SUM SUB MUL DIV

```
#include<iostream>
using namespace std;
int main() {
    float a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Sum            = " << a + b << endl;
    cout << "Subtraction    = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    if (b != 0)
        cout << "Division       = " << a / b << endl;
    else
        cout << "Division       = Infinity" << endl;
    return 0;
}
```

2a. LARGEST AMONG 3 NUMBERS

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c) cout << a << " is the largest." << endl;
    else if (b > c) cout << b << " is the largest." << endl;
    else cout << c << " is the largest." << endl;
    return 0;
}
```

2b. SMALLEST AMONG 3 NUMBERS

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a < b && a < c) cout << a << " is the smallest." << endl;
    else if (b < c) cout << b << " is the smallest." << endl;
    else cout << c << " is the smallest." << endl;
    return 0;
}
```

2c. LARGEST & SMALLEST (BOTH)

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c) cout << a << " is the largest." << endl;
    else if (b > c) cout << b << " is the largest." << endl;
    else cout << c << " is the largest." << endl;
    if (a < b && a < c) cout << a << " is the smallest." << endl;
    else if (b < c) cout << b << " is the smallest." << endl;
    else cout << c << " is the smallest." << endl;
    return 0;
}
```

3. AVERAGE OF 3 NUMBERS

```
#include<iostream>
using namespace std;
int main() {
    float a, b, c, avg;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    avg = (a + b + c) / 3;
    cout << "Average = " << avg << endl;
    return 0;
}
```

4a. RIGHT ANGLE TRIANGLE AREA

```
#include<iostream>
using namespace std;
int main() {
    float base, height, area;
    cout << "Enter base and height: ";
    cin >> base >> height;
    area = (base * height) / 2;
    cout << "Right Angle Triangle Area = " << area << endl;
    return 0;
}
```

4b. EQUILATERAL TRIANGLE AREA

```
#include<iostream>
using namespace std;
int main() {
    float a, area;
    cout << "Enter side length: ";
    cin >> a;
    area = (1.7320508f / 4) * a * a;
    cout << "Equilateral Triangle Area = " << area << endl;
    return 0;
}
```

4c. SCALENE TRIANGLE AREA

```
#include<iostream>
using namespace std;
int main() {
    float a, b, c, s, x, area;
    cout << "Enter sides a, b, c: ";
    cin >> a >> b >> c;
    s = (a + b + c) / 2;
    x = s * (s - a) * (s - b) * (s - c);
    float guess = x / 2;
    for (int i = 0; i < 20; i++) guess = (guess + x / guess) / 2;
    area = guess;
    cout << "Scalene Triangle Area = " << area << endl;
    return 0;
}
```

5. CIRCLE AREA & CIRCUMFERENCE

```
#include<iostream>
using namespace std;
int main() {
    float r, pi = 3.1416f, area, circ;
    cout << "Enter radius: ";
    cin >> r;
    area = pi * r * r;
    circ = 2 * pi * r;
    cout << "Area          = " << area << " Square Meter" << endl;
    cout << "Circumference = " << circ << " Meter" << endl;
    return 0;
}
```

6. POSITIVE OR NEGATIVE NUMBER

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n > 0) cout << n << " is Positive." << endl;
    else if (n < 0) cout << n << " is Negative." << endl;
    else cout << "The number is Zero." << endl;
    return 0;
}
```

7. ODD OR EVEN

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n % 2 == 0) cout << n << " is Even." << endl;
    else cout << n << " is Odd." << endl;
    return 0;
}
```

8. LEAP YEAR

```
#include<iostream>
using namespace std;
int main() {
    int year;
    cout << "Enter a year: ";
    cin >> year;
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        cout << year << " is a Leap Year." << endl;
    else
        cout << year << " is not a Leap Year." << endl;
    return 0;
}
```

9a. CELSIUS TO FAHRENHEIT

```
#include<iostream>
using namespace std;
int main() {
    float c, f;
    cout << "Enter temperature in Celsius: ";
    cin >> c;
    f = (c * 9 / 5) + 32;
    cout << c << " C = " << f << " F" << endl;
    return 0;
}
```

9b. FAHRENHEIT TO CELSIUS

```
#include<iostream>
using namespace std;
int main() {
    float f, c;
    cout << "Enter temperature in Fahrenheit: ";
    cin >> f;
    c = (f - 32) * 5 / 9;
    cout << f << " F = " << c << " C" << endl;
    return 0;
}
```

10a. UPPERCASE OR LOWERCASE (RANGE)

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 'a' && ch <= 'z') cout << ch << " is Lowercase." << endl;
    else if (ch >= 'A' && ch <= 'Z') cout << ch << " is Uppercase." << endl;
    else cout << ch << " is not a letter." << endl;
    return 0;
}
```

10b. UPPERCASE OR LOWERCASE (ASCII)

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 65 && ch <= 90) cout << ch << " is Uppercase. ASCII = " << (int)ch << endl;
    else if (ch >= 97 && ch <= 122) cout << ch << " is Lowercase. ASCII = " << (int)ch << endl;
    else cout << ch << " is not a letter." << endl;
    return 0;
}
```

11a. VOWEL OR CONSONANT - IF

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    if (ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'||ch=='A'||ch=='E'||ch=='I'||ch=='O'||ch=='U')
        cout << ch << " is a Vowel." << endl;
    else
        cout << ch << " is a Consonant." << endl;
    return 0;
}
```

11b. VOWEL OR CONSONANT - SWITCH

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << ch << " is a Vowel." << endl;
            break;
        default:
            cout << ch << " is a Consonant." << endl;
    }
    return 0;
}
```

12a. PRINT 10 TIMES - FOR LOOP

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) cout << "Bangladesh" << endl;
    return 0;
}
```

12b. PRINT 10 TIMES - WHILE LOOP

```
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 10) { cout << "Bangladesh" << endl; i++; }
    return 0;
}
```

12c. PRINT 10 TIMES - DO-WHILE

```
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    do { cout << "Bangladesh" << endl; i++; } while (i <= 10);
    return 0;
}
```

12d. PRINT TEXT N TIMES - FOR

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n;
    cout << "Enter text: ";
    cin.ignore(); getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    for (int i = 1; i <= n; i++) cout << text << endl;
    return 0;
}
```

12e. PRINT TEXT N TIMES - WHILE

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    cin.ignore(); getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    while (i <= n) { cout << text << endl; i++; }
    return 0;
}
```

12f. PRINT TEXT N TIMES - DO-WHILE

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    cin.ignore(); getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    do { cout << text << endl; i++; } while (i <= n);
    return 0;
}
```

13a. PRINT 1 TO 10

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) cout << i << " ";
    cout << endl;
    return 0;
}
```

13b. PRINT 1 TO N

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) cout << i << " ";
    cout << endl;
    return 0;
}
```

14a. SUM OF 1 TO 10

```
#include<iostream>
using namespace std;
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

14b. SUM OF 1 TO N

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

15. PRIME NUMBER CHECK

```
#include<iostream>
using namespace std;
int main() {
    int n, isPrime = 1;
    cout << "Enter a number: ";
    cin >> n;
    if (n < 2) isPrime = 0;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) { isPrime = 0; break; }
    }
    if (isPrime) cout << n << " is a Prime number." << endl;
    else cout << n << " is not a Prime number." << endl;
    return 0;
}
```

16. PERFECT NUMBER CHECK

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i < n; i++)
        if (n % i == 0) sum += i;
    if (sum == n) cout << n << " is a Perfect number." << endl;
    else cout << n << " is not a Perfect number." << endl;
    return 0;
}
```

17. SUM OF DIGITS

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        sum += temp % 10;
        temp /= 10;
    }
    cout << "Sum of digits of " << n << " = " << sum << endl;
    return 0;
}
```

18. REVERSE A NUMBER

```
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0;
    cout << "Enter a number: ";
    cin >> n;
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    cout << "Reversed number = " << rev << endl;
    return 0;
}
```

19. PALINDROME NUMBER

```
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        rev = rev * 10 + temp % 10;
        temp /= 10;
    }
    if (rev == n) cout << n << " is a Palindrome." << endl;
    else cout << n << " is not a Palindrome." << endl;
    return 0;
}
```

20. ARMSTRONG NUMBER

```
#include<iostream>
using namespace std;
int main() {
    int n, temp, digits = 0, sum = 0, t;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) { digits++; temp /= 10; }
    temp = n;
    while (temp != 0) {
        t = temp % 10;
        int p = 1;
        for (int i = 0; i < digits; i++) p *= t;
        sum += p;
        temp /= 10;
    }
    if (sum == n) cout << n << " is an Armstrong number." << endl;
    else cout << n << " is not an Armstrong number." << endl;
    return 0;
}
```

21. STRONG NUMBER

```
#include<iostream>
using namespace std;
int main() {
    int n, temp, sum = 0, digit, fact;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        digit = temp % 10;
        fact = 1;
        for (int i = 1; i <= digit; i++) fact *= i;
        sum += fact;
        temp /= 10;
    }
    if (sum == n) cout << n << " is a Strong number." << endl;
    else cout << n << " is not a Strong number." << endl;
    return 0;
}
```

22. GCD AND LCM

```
#include<iostream>
using namespace std;
int main() {
    int a, b, x, y, gcd, lcm;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    x = a; y = b;
    while (y != 0) { int t = y; y = x % y; x = t; }
    gcd = x;
    lcm = (a * b) / gcd;
    cout << "GCD = " << gcd << endl;
    cout << "LCM = " << lcm << endl;
    return 0;
}
```

23. FACTORIAL

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long fact = 1;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++) fact *= i;
    cout << n << "! = " << fact << endl;
    return 0;
}
```

24. SERIES: 1! + 2! + ... + n!

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact = 1;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        fact *= i;
        sum += fact;
    }
    cout << "1! + 2! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

25. SERIES: 2! + 4! + ... + n!

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (even number): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    cout << "2! + 4! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

26. SERIES: 1! + 3! + ... + n!

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (odd number): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    cout << "1! + 3! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

27a. FIBONACCI - PRINT TERMS

```
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
        c = a + b; cout << " " << c; a = b; b = c;
    }
    cout << endl;
    return 0;
}
```

27b. FIBONACCI - PRINT & SUM

```
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    long long sum = a + b;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
        c = a + b; cout << " " << c;
        sum += c; a = b; b = c;
    }
    cout << "\nSum = " << sum << endl;
    return 0;
}
```

28. SERIES: 1 + 2 + ... + n

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

29. SERIES: 1 + 3 + 5 + ... + n

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

30. SERIES: 2 + 4 + 6 + ... + n

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

31. SERIES: 1^1 + 2^2 + ... + n^n

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        long long p = 1;
        for (int j = 0; j < i; j++) p *= i;
        sum += p;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

32. SERIES: 1^2 + 2^2 + ... + n^2

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

33. SERIES: 1^2 + 3^2 + ... + n^2

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

34. SERIES: 2^2 + 4^2 + ... + n^2

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

35. SERIES: 1 + 2 + 4 + 8 + ...

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) { sum += term; term *= 2; }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

36. SERIES: 1 + 3 + 9 + 27 + ...

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) { sum += term; term *= 3; }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

37. SERIES: 1/1 + 1/2 + ... + 1/n

```
#include<iostream>
using namespace std;
int main() {
    int n; float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += 1.0f / i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

38. SERIES: 1/1^2 + 1/2^2 + ...

```
#include<iostream>
using namespace std;
int main() {
    int n; float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += 1.0f / ((float)i * i);
    cout << "Sum = " << sum << endl;
    return 0;
}
```

39. PATTERN - STARS INCREASING

```
*
**
***
****
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

40. PATTERN - NUMBERS INCREASING

```
1
12
123
1234
12345
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

41. PATTERN - REPEATED ROW NUM

```
1
22
333
4444
55555
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

42. PATTERN - 1 0 ALT START 1

```
1
10
101
1010
10101
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

43. PATTERN - 1 0 ALT ODD=1 EVEN=0

```
1
00
111
0000
11111
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

44. PATTERN - REPEATED ALPHABET

```
A
BB
CCC
DDDD
EEEEE
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

45. PATTERN - ALPHABETS INC

```
A
AB
ABC
ABCD
ABCDE
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

46. PATTERN - STARS DECREASING

```
*****
****
***
**
*
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

47. PATTERN - NUMBERS DECREASING

```
12345
1234
123
12
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

48. PATTERN - REP ROW DECREASING

```
55555
4444
333
22
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

49. PATTERN - 1 0 ALT DECREASING

```
10101
1010
101
10
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

50. PATTERN - ROW 1/0 DECREASING

```
11111
0000
111
00
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

51. PATTERN - ALPHABETS DEC

```
ABCDE
ABCD
ABC
AB
A
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

52. PATTERN - REP ALPHABET DEC

```
EEEEE
DDDD
CCC
BB
A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

53. PATTERN - STARS DIAMOND

```
*
**
***
**
*
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

54. PATTERN - NUMBERS DIAMOND

```
1
12
123
12
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

55. PATTERN - REP ROW NUM DIAMOND

```
1
22
333
22
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

56. PATTERN - 1/0 START 1 DIAMOND

```
1
10
101
10
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

57. PATTERN - ROW 1/0 DIAMOND

```
1
00
111
00
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

58. PATTERN - REP ALPHABET DIAMOND

```
A
BB
CCC
BB
A
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

59. PATTERN - ALPHABET SEQ DIAMOND

```
A
AB
ABC
AB
A
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

60. PATTERN - ALPHABET SEQ DIAMOND

```
/* Same as 59 */
```

```
// See program 59 above
```

61. PATTERN - RIGHT-ALIGNED STARS

```
    *
   **
  ***
 ****
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

62. PATTERN - RIGHT-ALIGNED NUMBERS

```
    1
   12
  123
 1234
12345
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

63. PATTERN - RIGHT-ALIGNED REP ROW

```
    1
   22
  333
 4444
55555
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

64. PATTERN - RIGHT-ALIGNED 1/0 START 1

```
    1
   10
  101
 1010
10101
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

65. PATTERN - RIGHT-ALIGNED ROW 1/0

```
    1
   00
  111
 0000
11111
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

66. PATTERN - RIGHT-ALIGNED REP ALPHA

```
    A
   BB
  CCC
 DDDD
EEEEE
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

67. PATTERN - RIGHT-ALIGNED ALPHA SEQ

```
    A
   AB
  ABC
 ABCD
ABCDE
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

68. PATTERN - LEFT-INDENT STARS DEC

```
****
 ***
  **
   *
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

69. PATTERN - LEFT-INDENT NUMBERS DEC

```
12345
 1234
  123
   12
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

70. PATTERN - LEFT-INDENT REP ROW DEC

```
55555
 4444
  333
   22
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

71. PATTERN - LEFT-INDENT 1/0 DEC

```
10101
 1010
  101
   10
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

72. PATTERN - LEFT-INDENT ROW 1/0 DEC

```
11111
 0000
  111
   00
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

73. PATTERN - LEFT-INDENT REP ALPHA DEC

```
EEEEE
 DDDD
  CCC
   BB
    A
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

74. PATTERN - LEFT-INDENT ALPHA SEQ DEC

```
ABCDE
 ABCD
  ABC
   AB
    A
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

75. PATTERN - SOLID STAR SQUARE

```
*****
*****
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

76. PATTERN - NUMBER SEQUENCE SQUARE

```
12345
12345
12345
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

77. PATTERN - ROW NUMBER SQUARE

```
1111
2222
3333
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

78. PATTERN - 1010 COL ALT SQUARE

```
1010
1010
1010
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

79. PATTERN - 1111/0000 ROW ALT SQUARE

```
1111
0000
1111
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= n; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

80. PATTERN - REPEATED ALPHABET SQUARE

```
AAAA
BBBB
CCCC
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

90. PATTERN - ALPHABET SEQ SQUARE

```
ABCD
ABCD
ABCD
```

```
#include<iostream>
using namespace std;
int main() {
    int n; cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

91. PATTERN - SEQUENTIAL NUMS TRIANGLE

```
1
2 3
4 5 6
```

```
#include<iostream>
using namespace std;
int main() {
    int n, count = 1;
    cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << count++ << " ";
        cout << endl;
    }
    return 0;
}
```

92. PATTERN - SEQUENTIAL FROM 0

```
0
1 2
3 4 5
```

```
#include<iostream>
using namespace std;
int main() {
    int n, count = 0;
    cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << count++ << " ";
        cout << endl;
    }
    return 0;
}
```

93. PATTERN - MULTIPLICATION TRIANGLE

```
1
2 4
4 6 9
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row * col << " ";
        cout << endl;
    }
    return 0;
}
```

94. PATTERN - PYRAMID OF STARS

```
  *
 ***
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: "; cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= (2 * row - 1); col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

95. FACTORIAL USING RECURSION

```
#include<iostream>
using namespace std;
long long fact(int n) {
    if (n == 0 || n == 1) return 1;
    return n * fact(n - 1);
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << n << "! = " << fact(n) << endl;
    return 0;
}
```

96. FIBONACCI USING RECURSION

```
#include<iostream>
using namespace std;
int fib(int n) {
    if (n <= 1) return n;
    return fib(n - 1) + fib(n - 2);
}
int main() {
    int n;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: ";
    for (int i = 0; i < n; i++) cout << fib(i) << " ";
    cout << endl;
    return 0;
}
```

97. CALCULATE POWER USING RECURSION

```
#include<iostream>
using namespace std;
long long power(int base, int exp) {
    if (exp == 0) return 1;
    return base * power(base, exp - 1);
}
int main() {
    int base, exp;
    cout << "Enter base and exponent: ";
    cin >> base >> exp;
    cout << base << "^" << exp << " = " << power(base, exp) << endl;
    return 0;
}
```

98a. ARRAY - FIND MAXIMUM

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max) max = arr[i];
    cout << "Maximum = " << max << endl;
    return 0;
}
```

98b. ARRAY - FIND MINIMUM

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int min = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] < min) min = arr[i];
    cout << "Minimum = " << min << endl;
    return 0;
}
```

99. ARRAY - FIND BOTH MAX AND MIN

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int max = arr[0], min = arr[0];
    for (int i = 1; i < n; i++) {
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
    }
    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    return 0;
}
```

100. SWAP TWO NUMBERS WITHOUT TEMP

```
#include<iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Before: a = " << a << ", b = " << b << endl;
    a = a + b;
    b = a - b;
    a = a - b;
    cout << "After : a = " << a << ", b = " << b << endl;
    return 0;
}
```

101a. CONTINUE EXAMPLE

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i % 3 == 0) continue;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

101b. BREAK EXAMPLE

```cpp
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 6) break;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```